DRONE DESIGN AT CATIA V5

SUMMER INTERNSHIP REPORT

submitted by

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in partial fulfilment for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

AEROSPACE ENGINEERING

Division of Aerospace Engineering

Karunya Institute of Technology and Sciences

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AEROSPACE & DEFENCE

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2025-07-10/Chennai, India.

CERTIFICATE / LOR

We are glad to inform you that **Majeesh S** from **Karunya institute of technology and sciences** has successfully completed the internship program at Brahmastra Aerospace & Defence Private Limited from **2025-04-04** to **2025-05-04**

During the internship, the candidate was exposed to various activities and discussions in the internship division, the candidate also closely worked as a part of the Project trainee program and the candidate successfully completed the role of project trainee entitled to **Drone design** with Catia V5 Brahmastra Aerospace & Defence Private Limited.

We found the candidate to be extremely inquisitive and hard-working. The candidate was very interested in learning the functions of our core division and willing to put their best efforts to understand the subject and willing to work towards creative thinking. The candidate worked with various software and met the objectives of the project.

The candidate's association with us was very fruitful and we wish him all the best in their future endeavors.

For Brahmastra Aerospace and Defence Pvt. Ltd.

Director.

Subash P Kuppusamy,

Founder & CEO

Objectives

Understanding Fundamental Design Principles**:** Gain a practical and deep understanding of the core engineering principles governing Unmanned Aerial Vehicle (UAV) design, including requirements analysis, structural integrity, and component integration.

Mastering CATIA V5 Proficiency**:** Acquire comprehensive technical skills and proficiency in the CATIA V5 Computer-Aided Design (CAD) software, utilizing its various workbenches (e.g., Part Design, Assembly Design, Drafting) for aerospace applications.

Detailed Component Modeling: Successfully conceptualize, model, and analyse the individual components of a drone system, ensuring design feasibility and dimensional accuracy using industry best practices in CAD.

System Integration and Assembly**:** Apply acquired CAD skills to create a cohesive and functional 3D assembly model of the complete drone, ensuring seamless integration of complex parts like the airframe, motors, and landing gear.

Applying Structured Methodology**:** Adhere to a professional, structured engineering approach to progress from a conceptual design brief to a fully detailed CAD model and technical documentation.

Effective Communication and Documentation**:** Synthesize the entire design process, findings, and technical data into a comprehensive report, effectively communicating the project's scope and outcomes, and meeting the objective of the project.

Methodology

Phase 1: Basics of Drone Systems and Design Requirements (Days 1–7)

The primary objective of this phase was to establish a solid theoretical foundation in Unmanned Aerial Systems (UAS) and define the scope of the design project.

- Theoretical Groundwork and Aerodynamics**:** Participants engaged in lectures and discussions covering the fundamental principles of flight dynamics and aerodynamics specific to drone platforms. This included studying different drone configurations (e.g., multi-rotor, fixed-wing) and their stability characteristics.

- System Component Analysis: A detailed examination of the major components comprising a modern drone system was conducted. This included the airframe, propulsion system, flight control system, and payload integration, allowing interns to discern their functions and interdependencies.
- Defining Design Constraint**s:** The phase culminated in defining the specific technical and operational requirements for the drone to be designed. This step was crucial for setting the groundwork and applying creative thinking to the design process.

Phase 2: Introduction and Application of CATIA V5 Fundamentals (Days 8–15)

The focus shifted to intensive training and hands-on application of the required CAD software, CATIA V5, a crucial tool for the project**.**

- CATIA V5 Familiarization**:** Interns underwent structured lectures and tutorials to familiarize themselves with the CATIA V5 software interface, navigation tools, and core workbenches.
- Part Design Mastery: Emphasis was placed on practical exercises in the Part Design module. Interns practiced foundational modeling techniques, including accurate 2D sketching, constraint application, and feature generation (e.g., Pad, Pocket, Shaft) to create preliminary 3D models of simple drone components**.**
- Best Practices and Efficiency**:** Interns gained insights into the best practices and techniques for efficient modelling, design iteration, and optimization within CATIA V5, preparing them to put their best efforts into the design work.

Phase 3: Detailed 3D Modeling and System Assembly (Days 16–23)

This core design phase involved translating the theoretical requirements into a complete, professional 3D CAD model, fulfilling the primary project role.

- Complex Component Modeling**:** Participants were tasked with conceptualizing and modeling the detailed, complex structures of the drone, such as the fuselage/airframe, motor mounts, and landing gear, applying advanced surfacing and solid modeling techniques in CATIA V5.
- System Assembly**:** Using the Assembly Design workbench, interns integrated all individual 3D models into a hierarchical and comprehensive assembly model of the

drone. Kinematic constraints were applied to simulate real-world component fits and movement, allowing them to meet the objectives of the project.

- Design Review and Optimization: Group discussions and peer review sessions provided opportunities to exchange ideas, troubleshoot modeling challenges, and enhance the understanding of CAD principles. Under the guidance of mentors, participants optimized their designs for structural soundness and material utilization.

Phase 4: Documentation and Final Synthesis (Days 24–30)

The culminating phase focused on the crucial task of report preparation and professional documentation, synthesizing the knowledge and skills acquired throughout the program.

- Data Synthesis and Analysis**:** Interns meticulously examined the design data, model structure, and assembly constraints, ensuring the final CAD project was robust and well-documented.
- Report Writing and Structuring**:** Participants synthesized their findings, observations, and design work into a coherent and informative report. Drawing upon their experience with various software, they documented the theoretical foundations, methodologies, design processes, and final model results.
- Effective Communication**:** Dedicated time was allocated for peer review sessions, where interns engaged in constructive feedback exchanges to enhance the clarity, rigor, and professionalism of their documentation, preparing them to communicate complex engineering concepts effectively.

Outcomes

Foundational and Technical Concepts:

The internship provided a thorough grounding in both UAV engineering principles and crucial CAD software skills, which are fundamental to the project:

- UAV Design Principles**:** The program commenced with the exploration of foundational concepts, including principles of UAV aerodynamics, structural requirements, and stability characteristics necessary for drone design. Through focused studies,

participants gained an appreciation for the factors governing flight performance and configuration choices.

- CAD Proficiency (CATIA V5)**:** Participants received comprehensive training in Computer-Aided Design (CAD) software, specifically CATIA V5, a crucial tool for professional aerospace engineering. Practical exercises and guided tutorials enabled the design and simulation of drone components, significantly enhancing proficiency in CAD modeling, surface design, and assembly integration.
- Comprehensive Understanding**:** The understanding achieved encompassed the theoretical knowledge of drone systems integrated with practical insights into the intricate process of digital design. Interns delved into concepts such as component optimization and assembly constraint management, enhancing their comprehension of the physics and engineering principles governing flight vehicle structure.

Practical Skills and Application:

The hands-on experience gained throughout the internship was instrumental in equipping participants with practical, industry-relevant skills:

- **Design and Modeling Proficiency:** Practical design projects and exercises honed skills in propulsion component design and CAD simulation. Participants successfully conceptualized, modelled, and iterated on drone components such as the airframe structure, motor mounts, and landing gear using CATIA V5**.**
- **Component Analysis and Optimization:** By applying theoretical knowledge to real-world design scenarios, interns developed the ability to assess design characteristics and optimize component geometry for performance and manufacturability. This practical experience enhanced the ability to translate conceptual ideas into detailed, validated designs.
- **Translating Concepts to Solutions**: Software training in CAD techniques, particularly CATIA V5, equipped interns with the tools necessary to translate theoretical concepts into tangible and practical design solutions, demonstrating an appreciation for design considerations in aerospace engineering**.**

Collaboration and Professional Growth:

Collaboration, mentorship, and effective communication were integral to the learning experience, fostering essential professional skills:

- Collaboration and Networking**:** Collaborative activities, including group discussions, problem-solving exercises, and design reviews, fostered an environment of mutual support and knowledge exchange. Interns cultivated teamwork, communication, and problem-solving skills, which are essential for success in the aerospace industry.
- Mentorship and Guidance: Mentorship from experienced professionals enriched the learning experience by providing valuable guidance, advice, and direction. This personalized attention empowered the intern to navigate challenges and gain insights into best practices and career development opportunities**.**
- Effective Communication**:** The dedicated time allocated for report preparation served as a crucial opportunity to refine communication and documentation skills. By synthesizing the design experiences and technical results into a comprehensive report, the intern enhanced the ability to articulate complex concepts clearly and present findings professionally to meet industry standards of clarity and rigor**.**

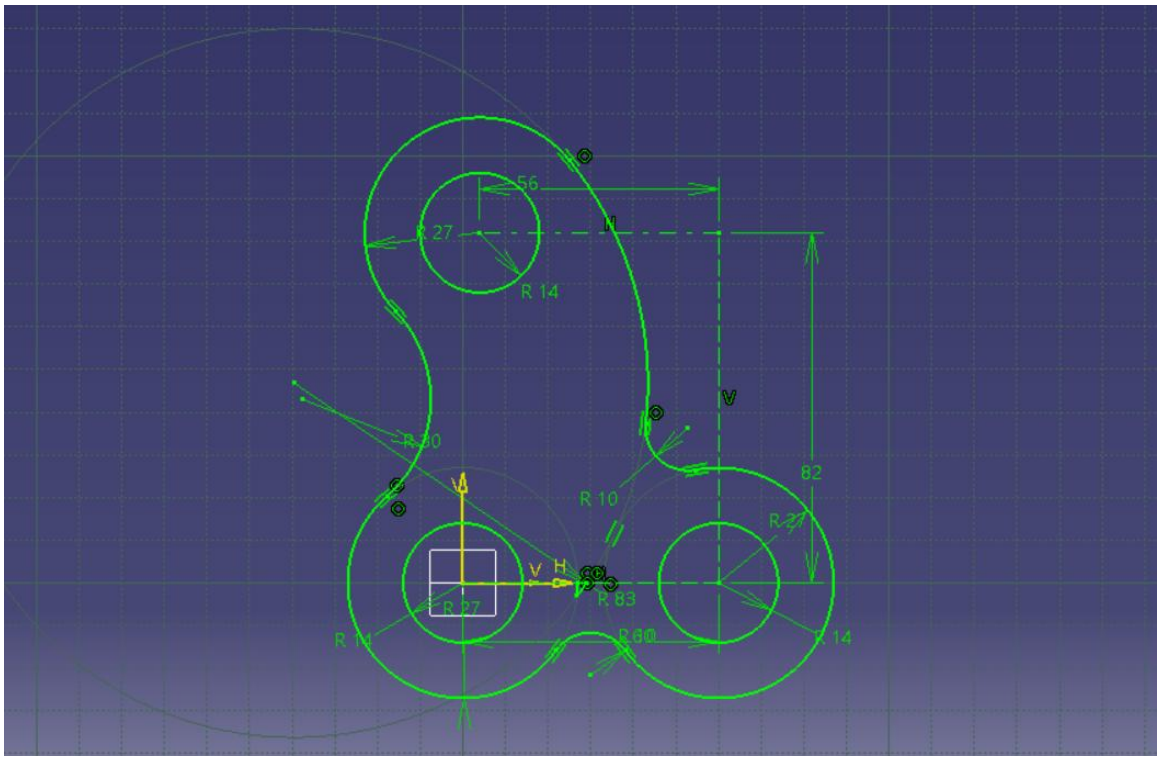
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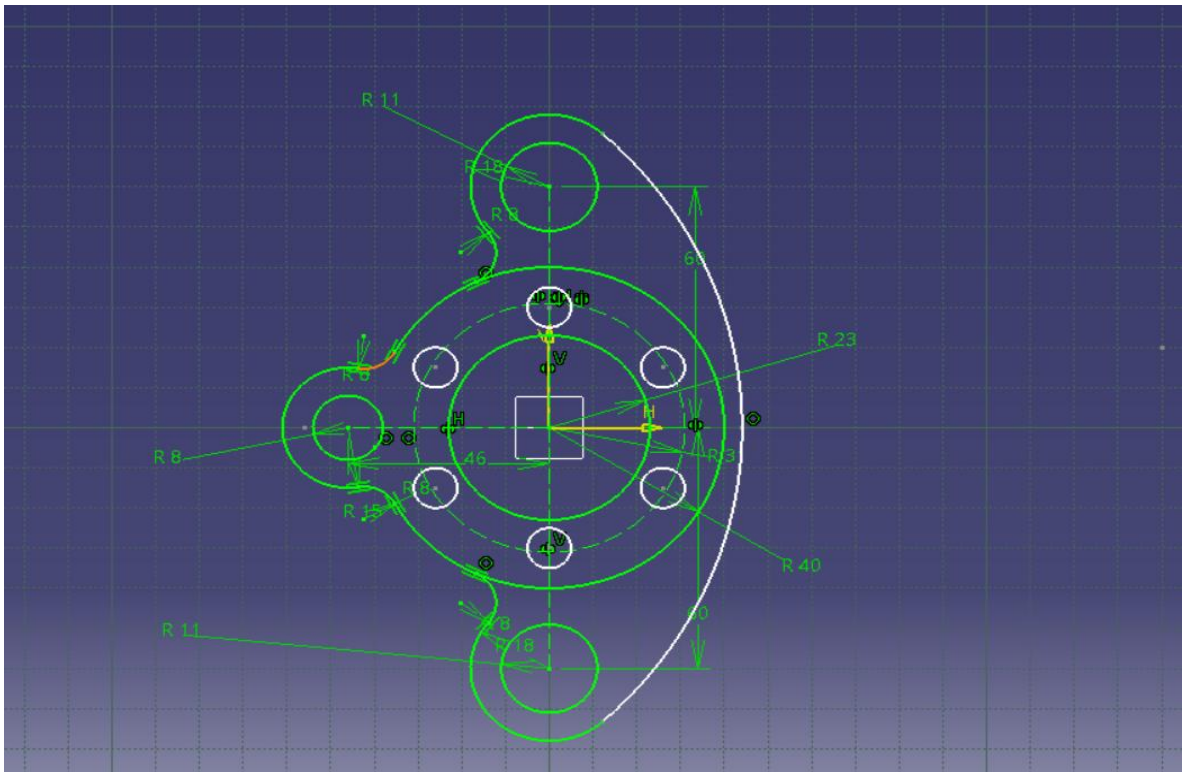
The internship served as a springboard, providing the confidence and skill set required to pursue future endeavours:

- Industry Readiness**:** Equipped with a comprehensive understanding of design principles, proficiency in CATIA V5, and essential soft skills, the intern is well-prepared to contribute effectively to aerospace engineering projects in industry settings, leveraging valuable hands-on experience.
- Advancing Academics**:** The solid foundation in CAD modeling and critical thinking skills prepares the intern for building advanced knowledge and expertise, supporting aspirations for further studies and research opportunities in specialized areas of aerospace engineering.
- Motivation for Innovation**:** The experience fostered a culture of curiosity and problem-solving, providing the confidence and motivation to pursue a passion for advancing the frontiers of aerospace technology and making meaningful contributions to the sector.

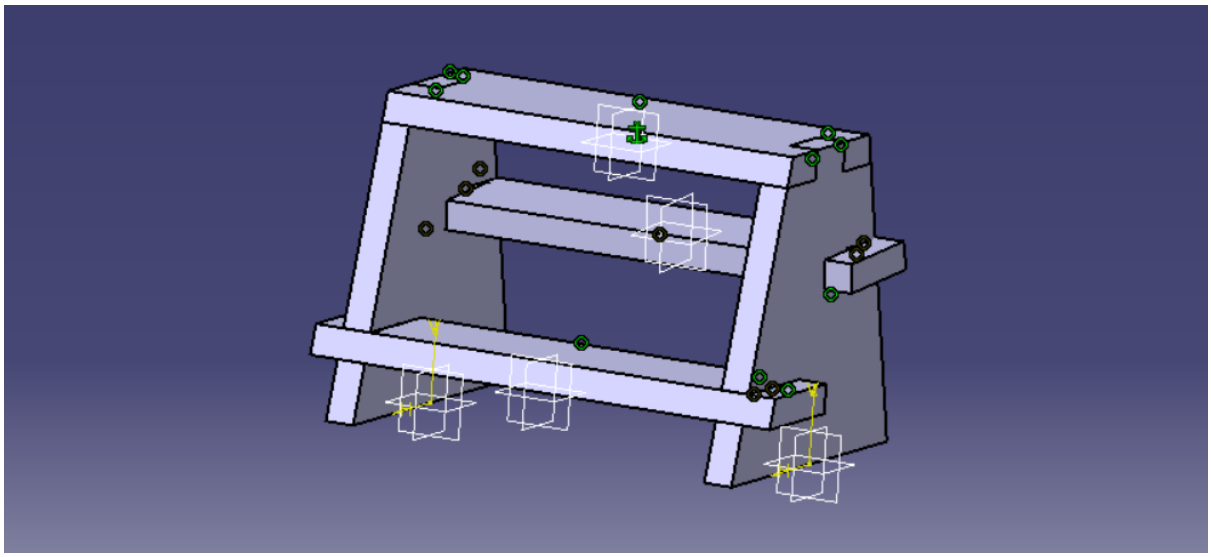
Project Design and Technical Outcomes

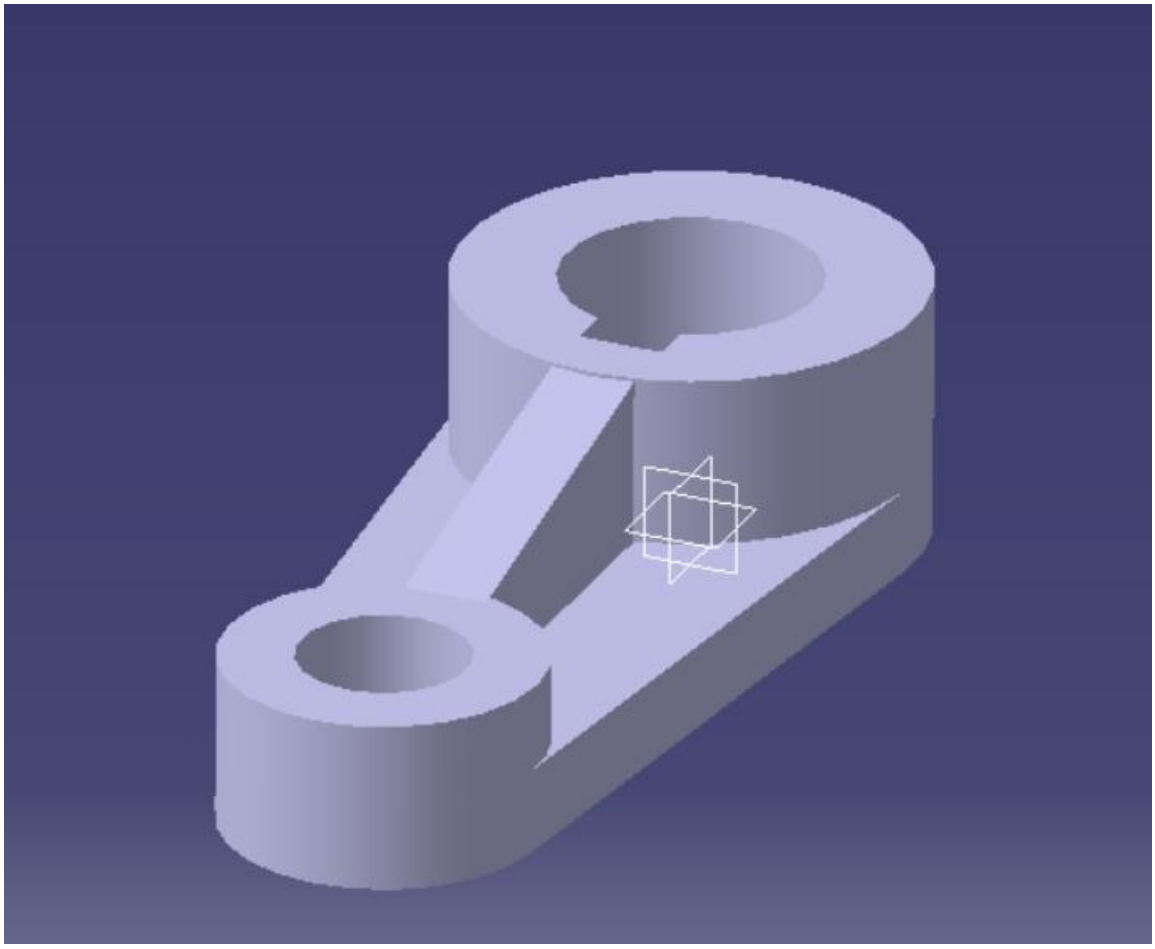
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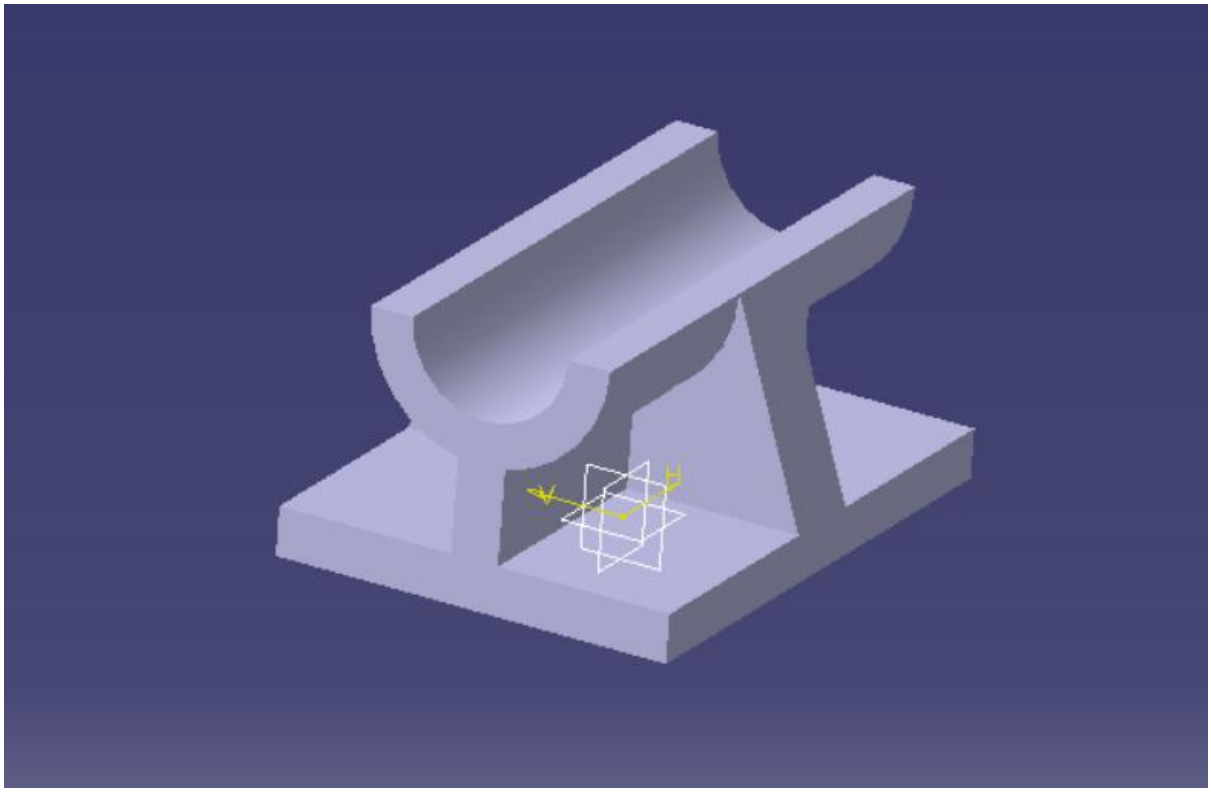


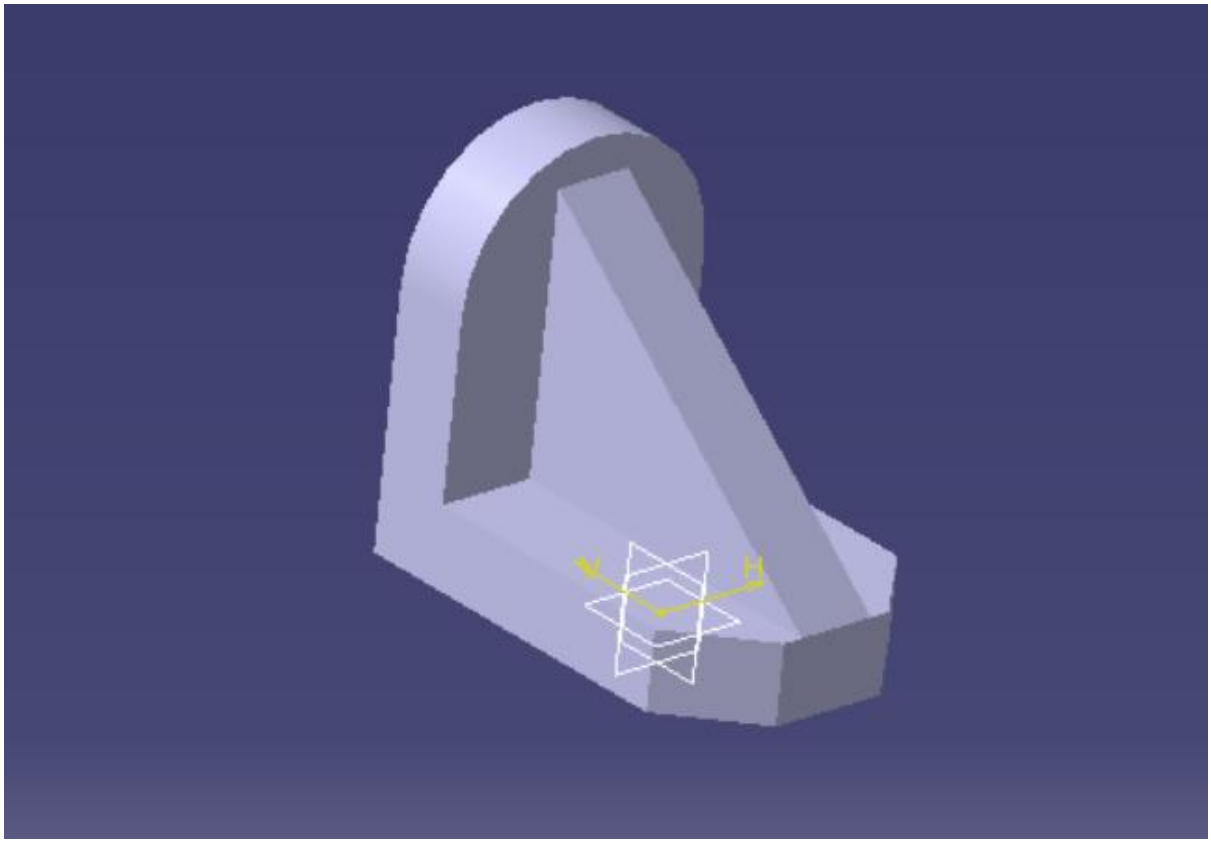


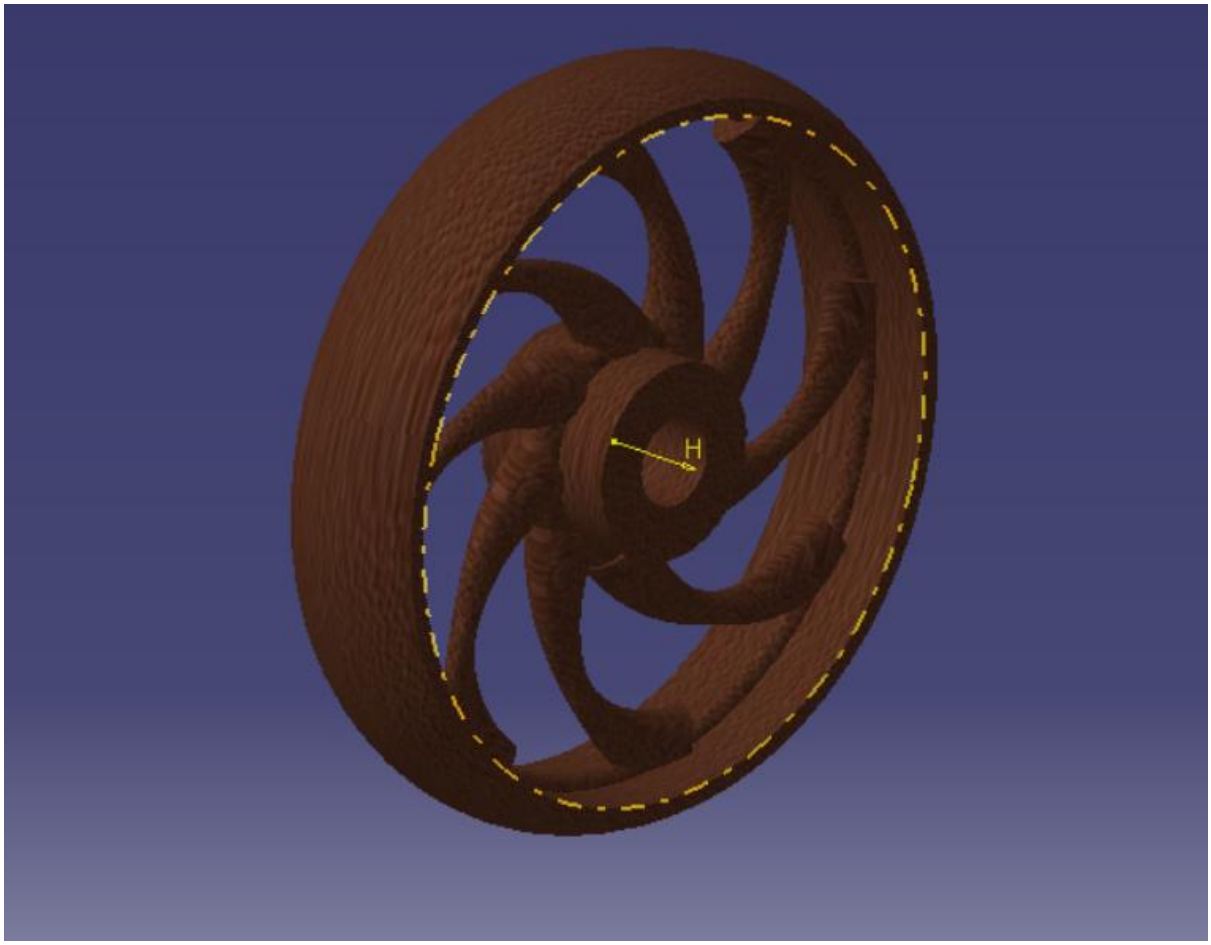


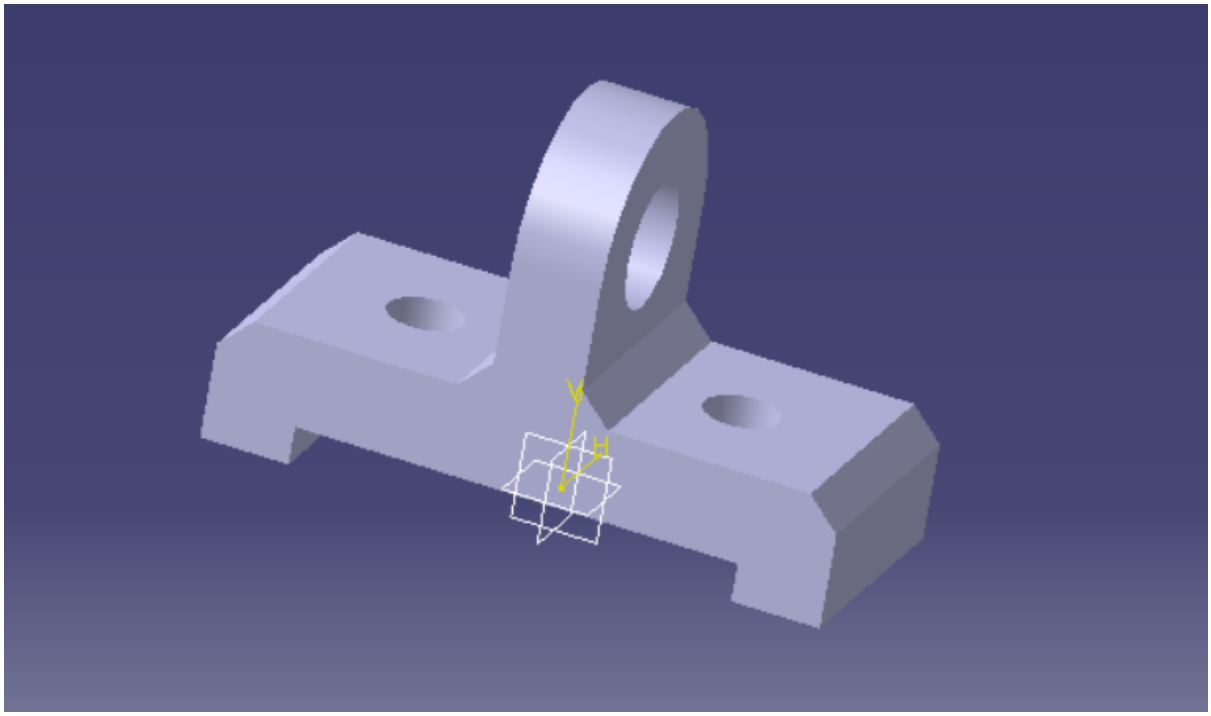


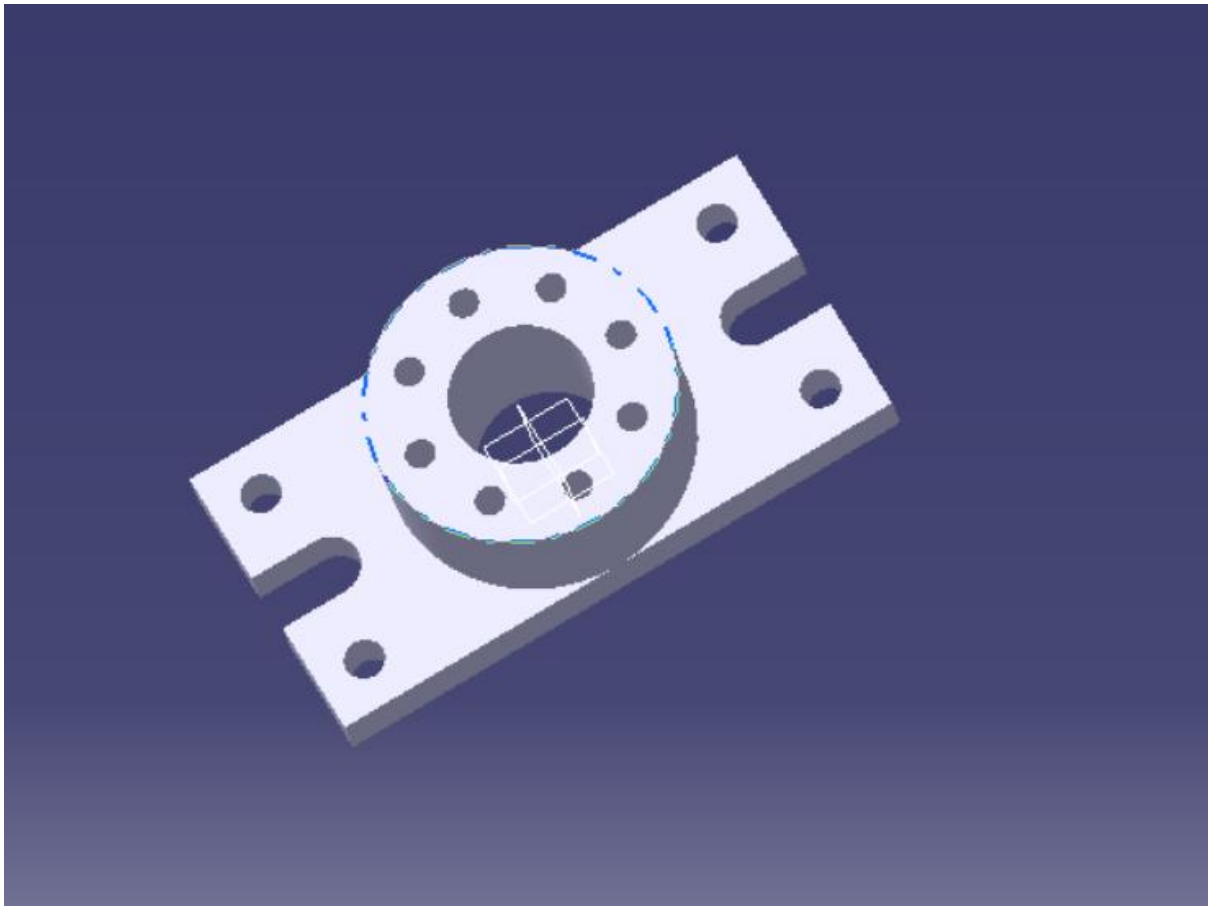


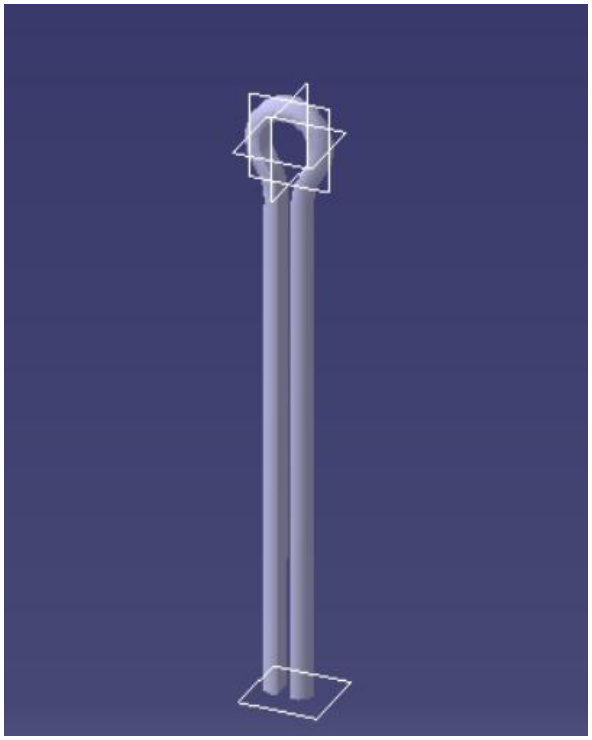


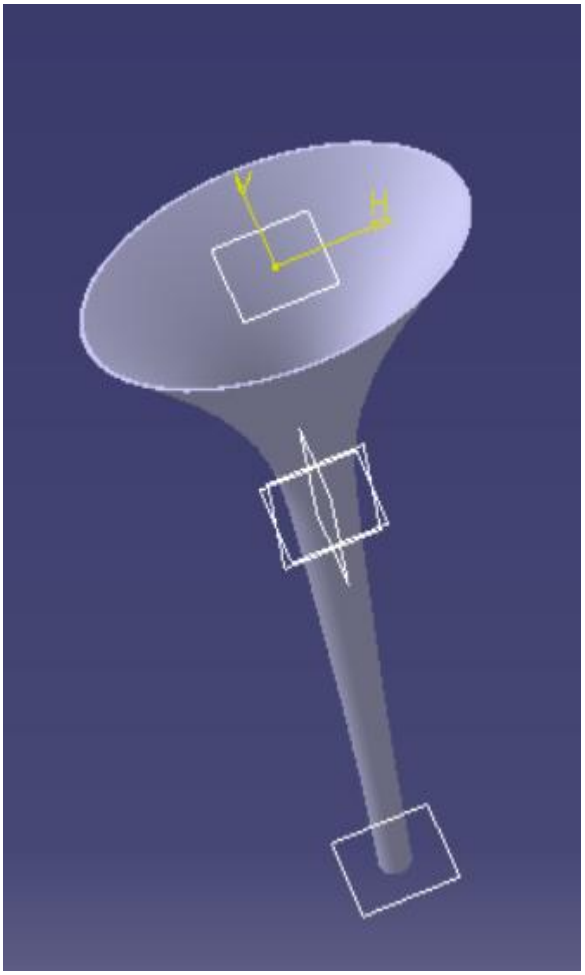

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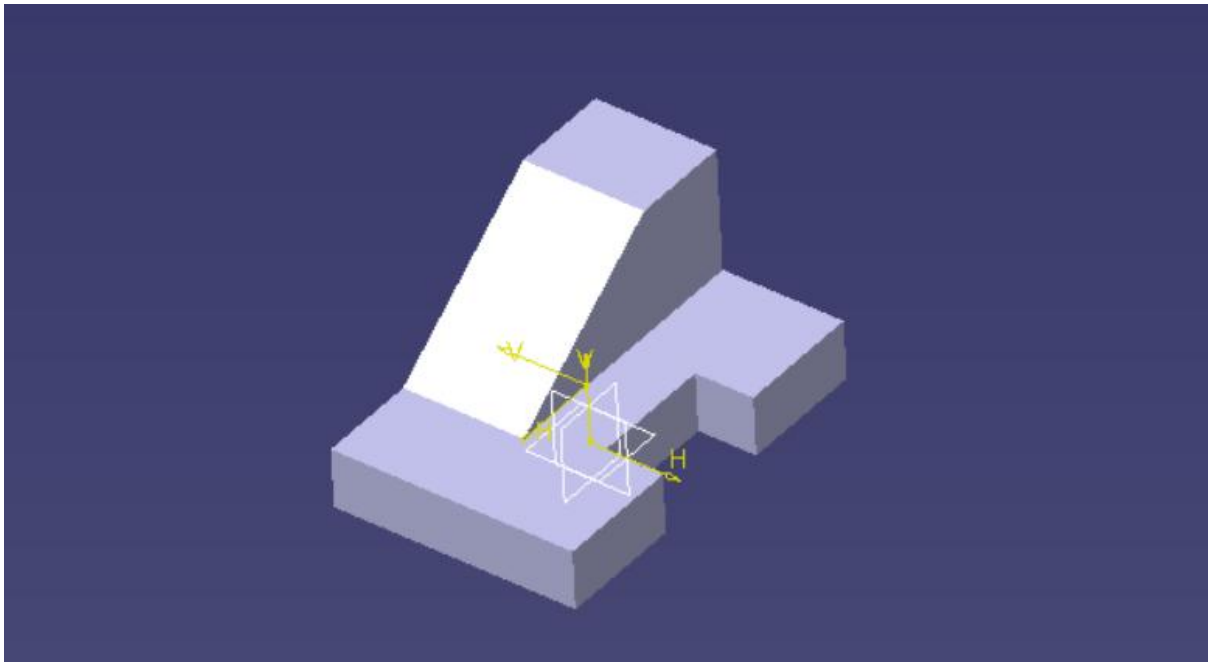

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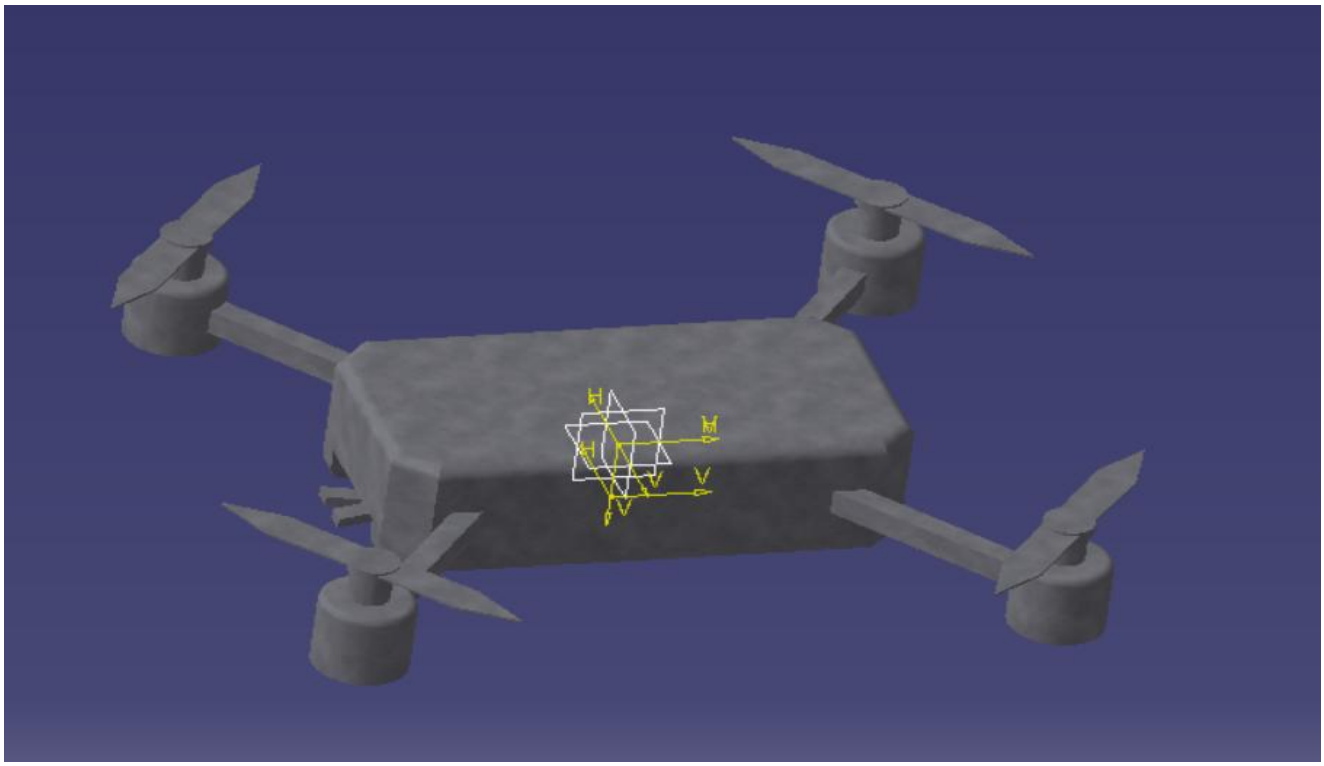


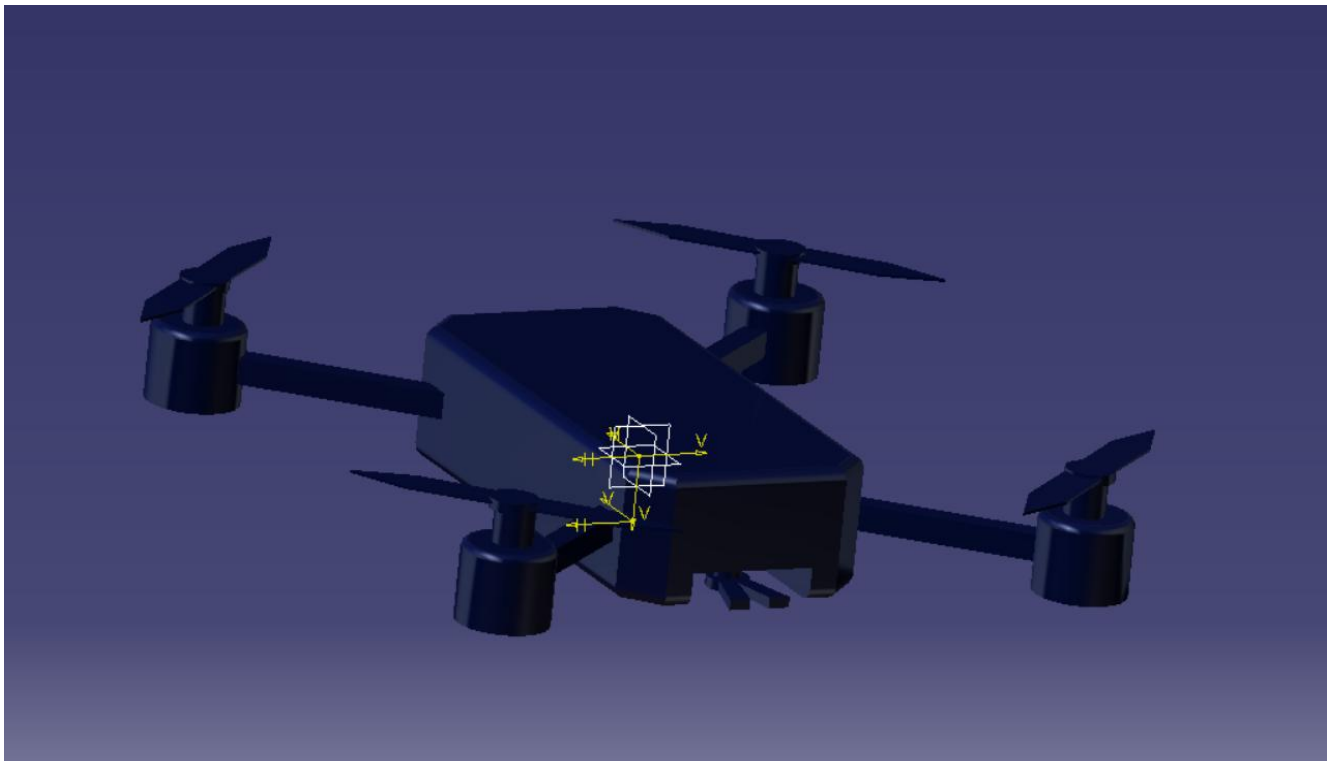


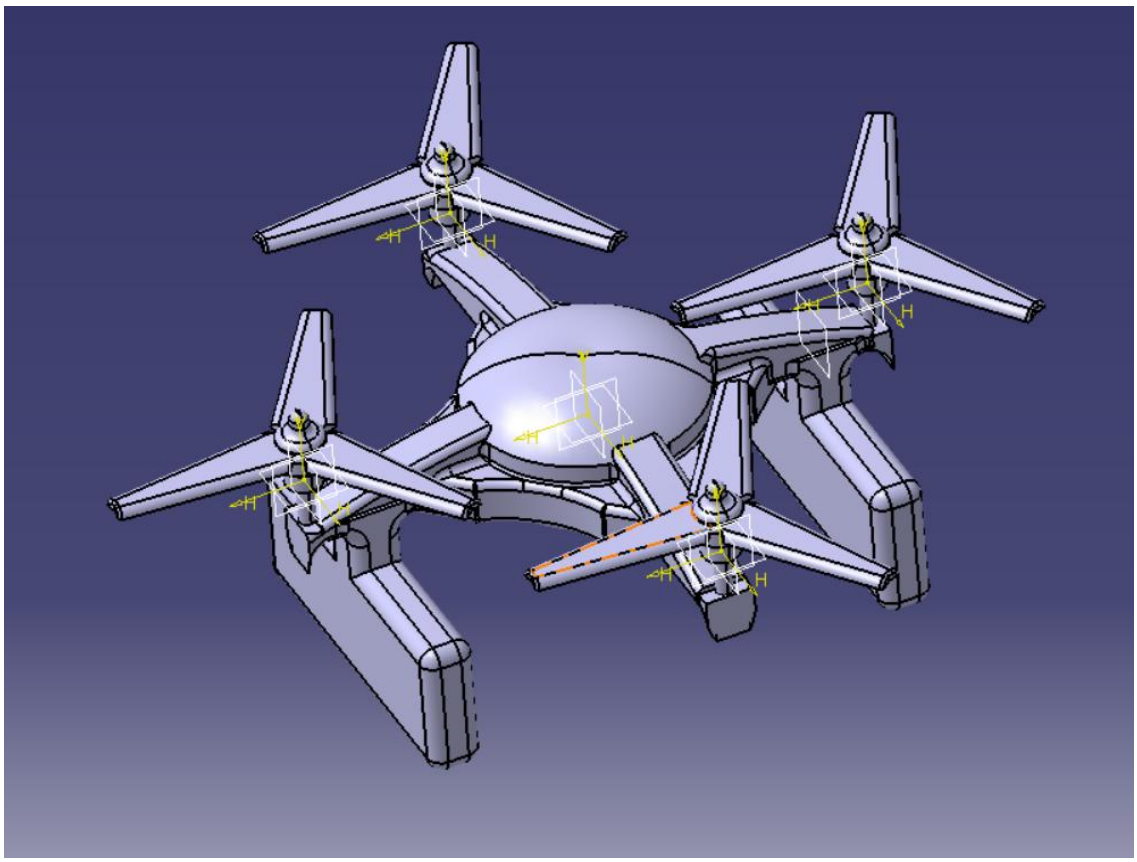

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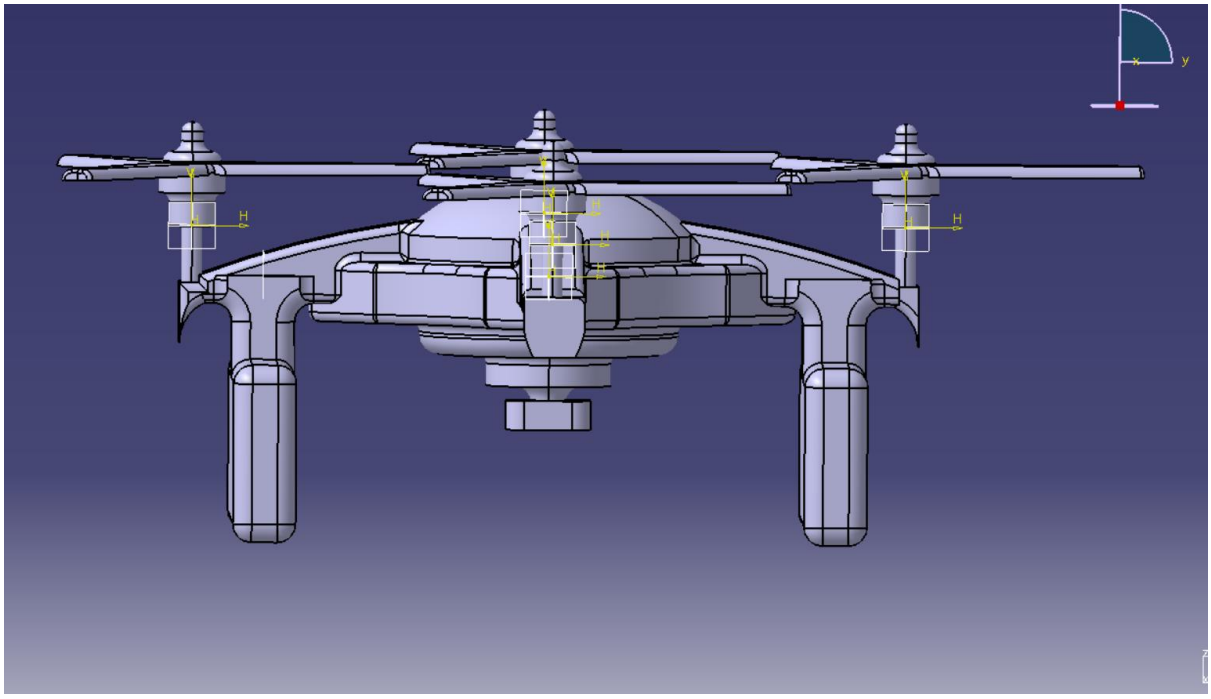

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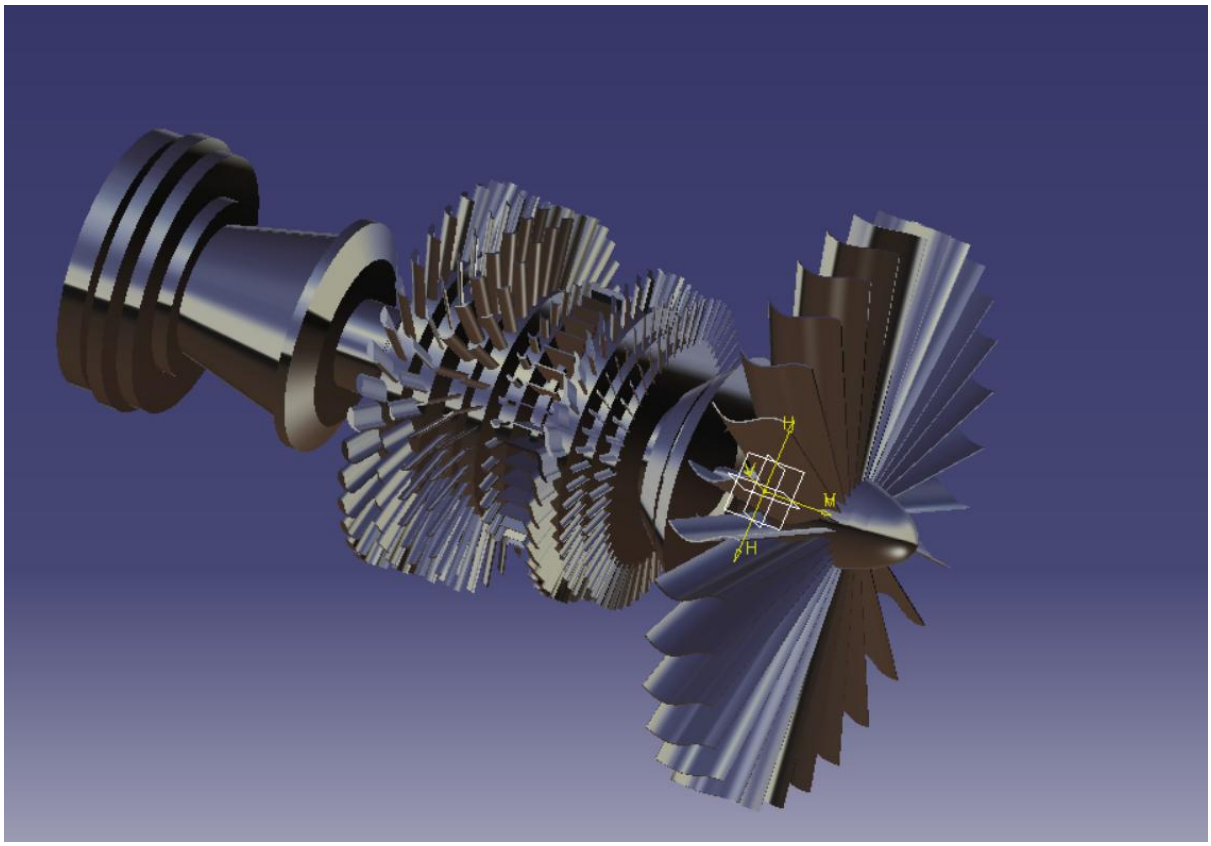


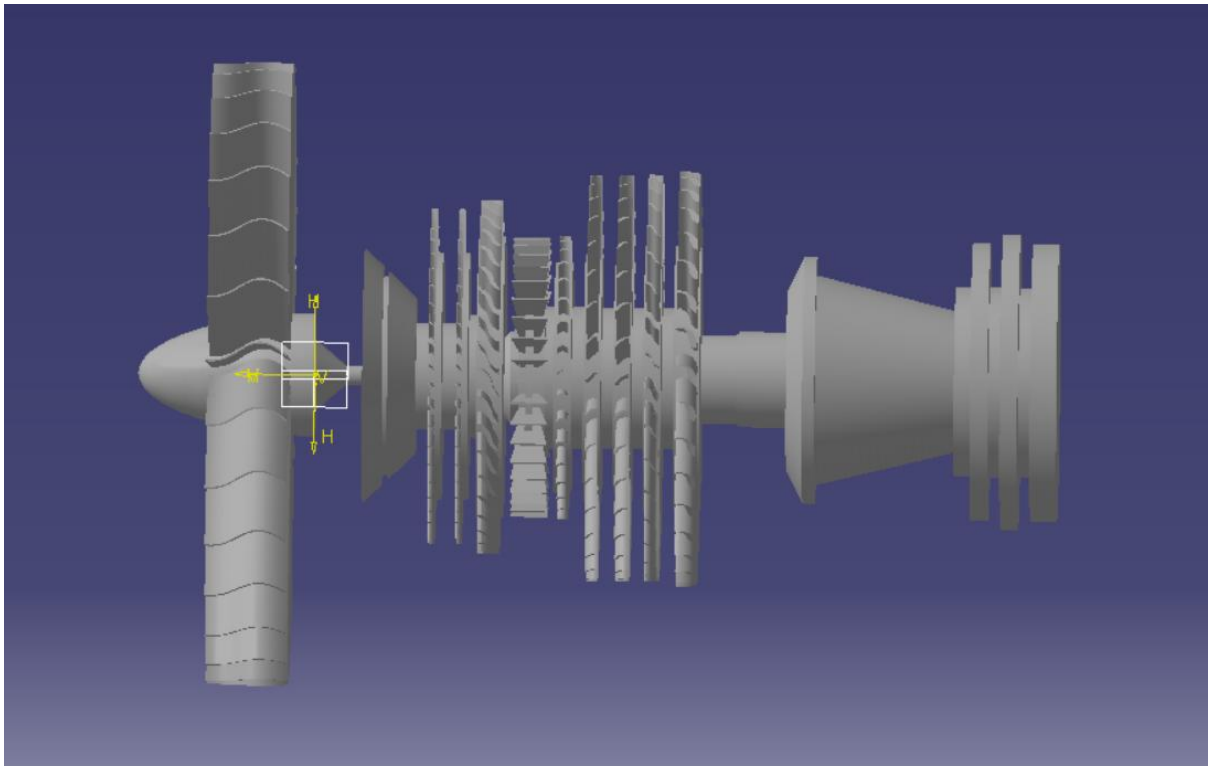






TURBOJET ENGINE:





Conclusion and Future Perspectives

The internship at Brahmastra Aerospace & Defence Private Limited, focused on the role of Project Trainee for **Drone design with Catia V5**, was a profound success, culminating in the complete conceptualization and detailed CAD modeling of a UAV system. This journey transformed theoretical knowledge into practical, industry-relevant design competence.

Summary of Achievements

The program successfully met all stated objectives, establishing a comprehensive skill set across technical and professional domains:

- Technical Proficiency in CATIA V5**:** The core outcome was the acquisition of comprehensive skills in **CATIA V5**, allowing for the conceptualization, modeling, and precise assembly of complex drone components. The successful digital realization of the quadcopter and other complex aerospace parts serves as tangible proof of acquired CAD proficiency.
- Application of Design Principles**:** Foundational knowledge of UAV aerodynamics, structural requirements, and component integration was effectively applied, translating design constraints into a functional 3D model.
- Professional Growth**:** Beyond technical skills, the experience fostered vital competencies in **teamwork, problem-solving, and effective technical documentation**. The intern was noted for being **extremely inquisitive and hard-working** and meeting the objectives of the project.

Future Endeavours

Equipped with a solid foundation in design principles, robust **CATIA V5** proficiency, and a professional network, this internship serves as a springboard for future academic and professional pursuits.

For future endeavours, the next logical steps for this project would involve:

1. Structural Analysis**:** Utilizing Finite Element Analysis (FEA) to simulate real-world stresses on the designed components, allowing for optimization of material usage and ensuring structural integrity.

2. Kinematic Simulation**:** Conducting motion studies on assembled components, such as the propeller rotation and potential landing gear mechanisms, to validate their operational freedom and function.